

NEDRE SOPPERO, Sweden

WMO: 020540

Lat: 68.0440N

Lon: 21.7550E

Elev: 360

StdP: 97.07

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -35.2	(c) -32.5	(d) -38.7	(e) 0.1	(f) -34.9	(g) -35.9	(h) 0.1	(i) -32.3	(j) 8.5	(k) -4.3	(l) 7.9	(m) -4.3	(n) 0.5	(o) 230	(p) 0.741

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	22.9	12.5	20.8	12.2	19.0	11.5	15.3	18.3	14.3	17.4	13.4	16.5	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	10.7	15.6	13.3	9.9	14.7	12.2	9.3	14.1	44.1	18.4	41.4	17.5	38.9	16.6	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.0	(b) 7.0	(c) 6.2	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 8.0	(b) 7.0	(c) 6.2	(d) WB	(e) -38.1	(f) 16.8	(g) 2.8	(h) 1.2	(i) -40.1	(j) 17.7	(k) -41.8	(l) 18.4	(m) -43.4	(n) 19.1	(o) -45.5	(p) 20.0	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-14.7	-14.4	-9.1	-2.3	4.6	10.8	13.6	11.4	6.4	-1.2	-8.3	-11.6
	DBStd	11.55	8.59	8.49	6.51	4.81	3.92	3.75	2.79	3.16	3.45	5.19	6.79	8.48
	HDD10.0	4327	767	683	591	369	175	35	4	22	118	346	548	670
	HDD18.3	7116	1025	916	849	619	426	228	147	216	359	604	798	928
	CDD10.0	254	0	0	0	0	7	57	117	65	8	0	0	0
	CDD18.3	3	0	0	0	0	0	1	2	0	0	0	0	0
	CDH23.3	36	0	0	0	0	5	14	14	3	0	0	0	0
	CDH26.7	2	0	0	0	0	0	2	0	0	0	0	0	0
(14) Wind	WSAvg	2.7	2.1	2.4	3.0	3.1	3.3	3.1	2.9	2.4	2.8	2.6	2.2	2.3
(15) Precipitation	PrecAvg	498	29	26	23	23	38	61	81	62	44	43	35	32
	PrecMax	624	52	61	71	56	87	117	161	112	93	89	73	93
	PrecMin	340	4	6	5	3	9	14	23	29	13	17	9	5
	PrecStd	82	11	15	16	15	20	26	33	22	24	20	17	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.8	6.6	10.9	23.8	25.0	25.1	23.8	16.3	10.7	5.7	4.5
		MCWB	0.3	0.7	2.3	4.5	10.7	13.8	14.2	13.0	9.6	6.0	3.2	1.3
	2%	DB	1.0	1.6	4.0	8.2	18.2	22.1	22.8	21.3	14.7	8.9	3.1	2.8
		MCWB	-1.0	-0.7	0.3	2.4	7.9	12.3	12.7	12.3	9.8	5.8	1.6	0.3
	5%	DB	-1.3	-0.9	2.2	6.5	14.3	19.7	20.7	19.5	13.3	7.4	1.3	0.5
		MCWB	-2.4	-2.3	-0.5	1.4	6.3	11.2	12.6	11.6	9.4	5.2	0.5	-0.8
	10%	DB	-3.4	-2.6	0.5	4.9	11.3	17.3	19.0	17.6	12.0	5.5	0.3	-0.7
		MCWB	-4.2	-3.6	-1.6	0.6	5.0	10.2	12.2	11.2	9.1	4.1	-0.3	-1.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.7	1.1	2.5	5.2	11.1	15.4	16.9	15.7	13.1	8.1	4.3	1.9
		MCDB	2.4	3.6	6.0	9.1	21.3	18.9	19.7	18.1	13.9	8.9	4.4	3.6
	2%	WB	-0.7	-0.5	0.9	3.2	9.0	14.3	15.8	14.5	11.6	6.6	1.7	0.5
		MCDB	0.7	1.2	2.9	6.9	15.7	18.5	18.8	17.8	12.8	7.9	2.7	1.8
	5%	WB	-2.4	-2.1	-0.3	1.9	7.4	12.8	14.9	13.6	10.8	5.2	0.7	-0.5
		MCDB	-1.4	-1.1	1.7	4.9	11.8	17.5	17.5	16.6	12.1	6.8	1.3	0.7
	10%	WB	-4.1	-3.6	-1.4	1.1	6.0	11.5	14.0	12.7	9.8	4.0	-0.2	-1.7
		MCDB	-3.3	-2.6	0.4	4.1	10.2	16.4	16.8	15.6	11.4	5.6	0.3	-0.7
(35) Mean Daily Temperature Range	MDBR		9.6	10.6	11.9	10.0	7.9	8.3	8.5	8.9	7.2	6.3	7.9	9.2
	5% DB	MCDBR	8.8	8.6	11.0	10.7	13.8	11.9	11.6	12.4	9.1	6.8	6.0	7.3
		MCWBR	8.1	7.6	8.3	5.8	5.5	4.6	4.0	4.9	5.2	4.4	5.2	6.7
	5% WB	MCDBR	8.2	8.5	10.0	8.9	10.9	10.4	8.1	9.8	6.1	6.0	5.2	7.1
		MCWBR	7.7	7.9	8.1	5.3	5.3	4.7	4.5	4.7	4.3	4.4	4.8	6.7
(40) Clear-Sky Solar Irradiance	taub		0.178	0.237	0.275	0.305	0.324	0.329	0.336	0.321	0.290	0.256	N/A	N/A
	taud		1.924	2.111	2.208	2.237	2.422	2.498	2.492	2.512	2.570	2.334	N/A	N/A
	Ebn at Noon		118	571	765	844	871	876	856	831	777	562	N/A	0
	Edh at Noon		8	46	76	100	94	90	88	77	56	37	N/A	0
(44) All-Sky Solar Radiation	RadAvg		0.04	0.49	1.99	3.93	4.98	4.98	4.66	3.35	1.82	0.60	0.09	0.00
	RadStd		0.00	0.04	0.09	0.17	0.45	0.37	0.44	0.26	0.21	0.09	0.01	0.00

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page